



KENYATTA UNIVERSITY
EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE

SMA 104: CALCULUS I

1ST SEMESTER 2023/2024

TIME: 2 HOURS

INSTRUCTIONS: Answer Question ONE and any other TWO Questions.

Question One [30 Marks] Compulsory

- (a) Let $f(x) = \frac{6-10x}{8x+7}$. Find $f^{-1}(x)$. [3 marks]
- (b) Let $f(x) = \sqrt{3x-4}$. Find $f'(x)$ using the definition of derivatives. [5 marks]
- (c) Find the derivative of y with respect to x where $y = \frac{e^x \cos x - \ln x}{x^3}$. [3 marks]
- (d) Find $\frac{dy}{dx}$ if $4x^2y^7 - 2x = x^5 + 4y^3$ [4 marks]
- (e) Find the equations of the tangent and normal lines to the curve $y^2 = 2x$ when $y = -4$ [3 marks]
- (f) A ladder 13m long has its foot on level ground and the top rests against a vertical wall. The foot begins to slip away, from the wall at the rate of 0.25m/min. At what rate is the top slipping down the wall when the foot of the ladder is 5m away from the wall [5 marks]
- (g) Determine the nature of all the stationary points on the curve $y = x^3 - 3x^2 - 9x + 1$ [3 marks]
- (h) Use differentials to approximate $\cos 43^\circ$ [4 marks]

Question Two [20 Marks]

- (a) Given that $f(2) = 3, g(3) = 2, f(3) = 4$ and $g(2) = 5$. Find

- (i) $(f \circ g)(3)$
(ii) $(g \circ f)(2)$

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- (b) Evaluate $\lim_{x \rightarrow \infty} \frac{\sqrt{3x^2+6}}{5-2x}$

- (c) Differentiate $y = \frac{1}{\sqrt{x+2}}$ from first principles [4 marks]
- (d) Let $f(x) = \sqrt{x^2-1}$ and $g(x) = 3x^2+1$. Evaluate $(f^{-1} \circ g^{-1})(x)$. [4 marks]
- (e) Determine the values of the constants α and β which make the following function continuous on all values of x

$$f(x) = \begin{cases} \alpha x - \beta, & \text{for } x \leq -1 \\ 2x^2 + 2\alpha x + \beta, & \text{for } -1 < x \leq 1 \\ 4, & \text{for } x > 1 \end{cases}$$

Question Three [20marks]

(a) Given that $y = \ln \sqrt{\frac{1-x^2}{1+x^2}}$, show that $\frac{dy}{dx} = \frac{-2x}{1-x^2}$ [5 marks]

(b) Differentiate $y = \frac{x^2 \sin(3x^2-1)}{e^{4x} + \ln x}$ with respect to x [5 marks]

c] Differentiate the following functions with respect to x

i. $y = e^{2x} \ln(5x+8)$ [3 marks]

ii. $\sin y + x^2 y^3 - \cos x = 2y$ [3 marks]

d] Find y' given that $y = \sin^{-1}(\cos x)$ [4 marks]

Question Four [20 marks]

(a) Find $\frac{dy}{dx}$ where $\tan(x^2 y^4) = 3x + y^2$ [5 marks]

(b) Let $x = 4 \sin t$ and $y = 12 \cos t$. Find

(i) $\frac{dy}{dx}$ [4 marks]

(ii) $\frac{d^2 y}{dx^2}$ [3 marks]

(c) Find the equation of the tangent and normal to the curve $3x^2 - y + 1 = y^2 - 2$ at point $(-1, 0)$ [8 marks]

Question Five [20 marks]

(a) Find all the asymptotes, if any, of the curve $y = \frac{x^2 + 4x + 7}{x-1}$ [5 marks]

(b) Use differentials (small changes) to estimate the value of $\sqrt{24.6}$ to 2 decimal places. [5 marks]

(c) The side of a cube is increasing at the rate of 5 cm/s^{-1} . Find the rate of increase of the volume when the length is 3cm. [4 marks]

(d) A bullet is shot vertically upwards and its height, h meters, at time t seconds is given by $h = -3.2t^2 + 256$

(i) Find the velocity with which the bullet was shot [2 marks]

(ii) How fast is the bullet moving at $t = 15$ seconds [2 marks]

(iii) How high does the bullet travel? [2 marks]

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KENYATTA UNIVERSITY

EXAMINATION FOR THE DEGREE OF BACHELOR OF EDUCATION (SCIENCE),
BACHELOR OF EDUCATION (ARTS)

SMA 114: CALCULUS 1

2ND SEMESTER 2023/2024

TIME: 2 HOURS

INSTRUCTIONS: Answer Question ONE and any other TWO Questions.

QUESTION ONE (30 MARKS)

a) Evaluate the following limits

i. $\lim_{x \rightarrow -4} \left(\frac{2x+8}{x^2+x-12} \right)$ (3 marks)

ii. $\lim_{x \rightarrow \infty} \sqrt[3]{\left(\frac{3x+5}{6x-8} \right)}$ (3 marks)

b) Test the continuity of the following function at $x = 3$

$$f(x) = \begin{cases} \frac{1}{x+2}, & x < -2 \\ x^2 - 5 & -2 < x \leq 3 \\ \sqrt{x+13} & x > 3 \end{cases}$$
 (4 marks)

c) Find the derivative of the following functions.

i. $y = (x^2 + 1)^{\cos x}$ (3 marks)

ii. $y = e^{2x} \ln(5x + 8)$ (2 marks)

d) Given the following functions $f(x) = 5x^2 - 3x + 20$ and $g(x) = 2x^2 + 3$. Find $g^{-1}(f(x))$ (4 marks)

e) A function $f: \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(x) = x^2$. State and justify if the function is

i. One-to-one. (1Mark)

ii. Many-one (1Mark)

iii. Onto (1Mark)

f) Given that $x = \cos 5t$, $y = \sin 5t$ find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ (4 marks)

g) Use the equation $y = \sqrt{x} + \frac{1}{\sqrt{x}}$ to approximate $\sqrt{102}$ (4 marks)

QUESTION TWO (15 MARKS)

a) Evaluate the following limits

i. $\lim_{x \rightarrow 7} \frac{\sqrt{x+2} - 3}{x-7}$ (3 marks)

ii. $\lim_{x \rightarrow 0} \frac{\sin 2x}{x \cos 3x}$ (4 marks)

b) Find the value of the constant k so that the function $f(x)$ is continuous for all x

$$f(x) = \begin{cases} 2x + k, & \text{if } x \leq 3 \\ 2k - x, & \text{if } x > 3 \end{cases} \quad (4 \text{ marks})$$

c) Find the domain of the function

$$f(x) = \begin{cases} -x - 1 & \text{if } x < -1 \\ 0 & \text{if } -1 \leq x \leq 1 \\ x + 1 & \text{if } x > 1 \end{cases} \quad (1 \text{ marks})$$

d) Given the following functions $f(x) = 10x^2 - 6x + 7$ and $g(x) = 2x + 1$. Find the composite functions. $f(g(x))$ (3 marks)

QUESTION THREE (15 MARKS)

a) By simplifying completely, find $\frac{dy}{dx}$ given that $y = 2\sqrt{x} \tan^{-1}(\sqrt{x})$ [3 marks]

b) Find the equation of the tangent to the curve $y = \sinh x + \cosh x$ at the point P(0,1).
Leave your answer in the form $y = mx + c$. [3 marks]

c) If $y = x^{\sqrt{x}}$, show by strictly using logarithmic differentiation that:

$$\frac{dy}{dx} = x^{\sqrt{x}} \left(\frac{2 + \ln x}{2\sqrt{x}} \right).$$

[3 marks]

d) Find $\frac{dy}{dx}$ given that $\cos(x - y) = xe^x$.

[3 marks]

e) By using parametric differentiation and given the parametric equations:

$x = 3 \sec t$ and $y = 18 \tan t - 5$, show that the expression for $\frac{dy}{dx}$ in terms of t is given by

$$\frac{dy}{dx} = 6 \csc t.$$

[3 marks]

QUESTION FOUR (15 MARKS)

a) A box with a square base and open top must have a volume of $32,000 \text{ cm}^3$. Find the dimensions of the box that minimize the amount of material used. [7 marks]

b) The curve C has equation $y = \frac{1}{2}x^3 - 9x^{\frac{3}{2}} + \frac{8}{x} + 30, x > 0$. Find an equation of the normal to C at the point $P(4, -8)$ giving your answer in the form $ax + by + c = 0$ where a, b and c are integers. [4 marks]

c) Oil is spilling from a ruptured tanker and spreads in a circular pattern. If the radius of the oil spill increases at a constant rate of 1 metre/second , how fast is the area of the spill increasing when the radius is 30 metres ? [4 marks]

QUESTIONS FIVE (15 MARKS)

A particle moves according to a law of motion $s = t^3 - 12t^2 + 36t, t \geq 0$, where t is measured in seconds and s in metres.

a) Find the velocity at time t . [2 marks]

b) What is the velocity in m/s after 3 seconds? [2 marks]

c) When is the particle at rest? [3 marks]

d) When is the particle moving forward (that is in the positive direction)? [3 marks]

e) Find the total distance travelled during the first 8 seconds. [2 marks]

f) Find the acceleration in m/s^2 at time t and after 3 seconds. [3 marks]

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EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE

SMA 104: CALCULUS 1

1ST SEMESTER 2024/2025

TIME: 2 HOURS

INSTRUCTIONS: Answer Questions ONE and any other TWO.

QUESTION ONE (30 Marks)

- a). Suppose that $f(x) = \frac{1}{x+2}$ and $g(x) = \frac{4}{x-1}$. Find the following composite function $f \circ g$ [2 marks]
- b). Obtain the inverse of $f(x) = \frac{2x+3}{x+4}$ [4 mark]
- c) Given that $g(x) = \begin{cases} x^3 - 2x, & \text{if } x \geq 4 \\ x+1, & \text{if } x < 4 \end{cases}$, evaluate $g(6)$, $g(2)$ and $g(-2)$ [3 marks]
- d) Is there a number a such that $\lim_{x \rightarrow -2} \left(\frac{ax^2 + 15x + 15 + a}{x^2 + x - 2} \right)$ exists? If so, find the value of a and the value of the limit. [4 marks]
- e) Find, from first principles, the derivative of $f(x) = \frac{1}{2x}$ [4 marks]
- f) Using appropriate technique, find $\frac{dy}{dx}$, given that $y = \frac{2x^2 + 1}{x^2 - 1}$ [4 marks]
- g) Given the function $f(x) = \begin{cases} c - x & \text{if } x \leq \pi \\ c \sin x & \text{if } x > \pi \end{cases}$
- i. Find the value of the constant c so that the function $f(x)$ is continuous. [4 marks]
- ii. For the value of c found in part (i), verify whether the three conditions for continuity are satisfied [5 marks]

QUESTION TWO (20 Marks)

- a) Use the definition of the derivative to find $f'(2)$ for the function $f(x) = x + \frac{1}{x}$
[6 marks]
- b) Given that, $y = (x^2 + 3x)^x$ find $\frac{dy}{dx}$
[4 marks]
- c) Differentiate the function $f(x) = \sin x \cos 2x$
[4 marks]
- d) A ladder 12 metres long leans against a wall. The foot of the ladder is pulled away from the wall at the rate of $\frac{1}{2}$ metres per minute. At what rate is the top of the ladder falling when the foot of the ladder is 4 metres from the wall?
[6 marks]

QUESTION THREE (20 MARKS)

- a. Given the following functions $f(x) = \sqrt{2x^2 - 4}$ and $g(x) = 4x^2 - 2$. Find $f^{-1}(g^{-1}(x))$
(6 marks)
- b. Differentiate the following functions with respect to x
- i. $y = (x^2 + 1)^{\sin x}$
(5 marks)
- ii. $y = \sin^{-1}(2x) \ln(\cos x)$
(5 marks)
- iii. $y = (3x - 4)^5 (7x + 6)$
(4 marks)

QUESTION FOUR (20 MARKS)

- a) Evaluate the following limits
- i. $\lim_{x \rightarrow 4} \left(\frac{2x + 8}{x^2 + x - 12} \right)$
(3 marks)
- ii. $\lim_{x \rightarrow \infty} \left(\frac{3x^2 + 8x - 4}{2x^2 + 4x - 5} \right)$
(3 marks)
- iii. $\lim_{x \rightarrow 0} \frac{\sin 4x}{x}$
(3 marks)
- b) Distinguish the stationary points of the curve and sketch the curve $y = x^3 - 3x^2 - 9x + 25$.
(6 Marks)

- c) Find the equation of the tangent and normal to $x^2y^3 + 6x^2 - y = 12$ at the point (1,2) on the curve. (5 marks)

QUESTION FIVE (20 MARKS)

- a) Evaluate Given that $x = 1 + t^3$, $y = 2t^2 - 4$ find $\frac{d^2y}{dx^2}$ (3 marks)
- b) Differentiate the function $f(x) = \cos x$ using first principles (5 marks)
- c) An open box is to be made from a 16 inch by 30 inch piece of card board by cutting out squares of equal size from the four corners and bending up the sides. What size should the squares be to obtain a box with the largest volume? (7 marks)
- (d) Use the equation $y = \sqrt{x}$ to approximate $\sqrt{99.7}$. (5 marks)

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